Food Delivery Time Predictor

In [2]:
```
# import the libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
```

In [3]:
```
# Dataset
data = {
    'distance_km': [2.5, 6.0, 1.2, 8.5, 3.8, 5.2, 1.8, 7.0, 4.5, 9.2,
                    2.0, 6.5, 3.2, 7.8, 4.0, 5.8, 1.5, 8.0, 3.5, 6.8,
                    2.2, 5.5, 4.2, 9.0, 2.8, 7.2, 3.0, 6.2, 4.8, 8.2],

    'prep_time_min': [10, 20, 8, 25, 12, 18, 7, 22, 15, 28,
                      9, 19, 11, 24, 14, 17, 6, 26, 13, 21,
                      10, 16, 14, 27, 11, 23, 12, 18, 15, 25],

    'delivery_time_min': [18, 38, 12, 52, 24, 34, 14, 45, 29, 58,
                          15, 40, 21, 50, 27, 35, 11, 54, 23, 43,
                          17, 32, 26, 56, 19, 47, 20, 37, 30, 53]
}
```

In [4]:
```
df = pd.DataFrame(data)
```

In [5]:
```
df.head()
```

Out[5]:

	distance_km	prep_time_min	delivery_time_min
0	2.5	10	18
1	6.0	20	38
2	1.2	8	12
3	8.5	25	52
4	3.8	12	24

In [6]:
```
df.describe()
```

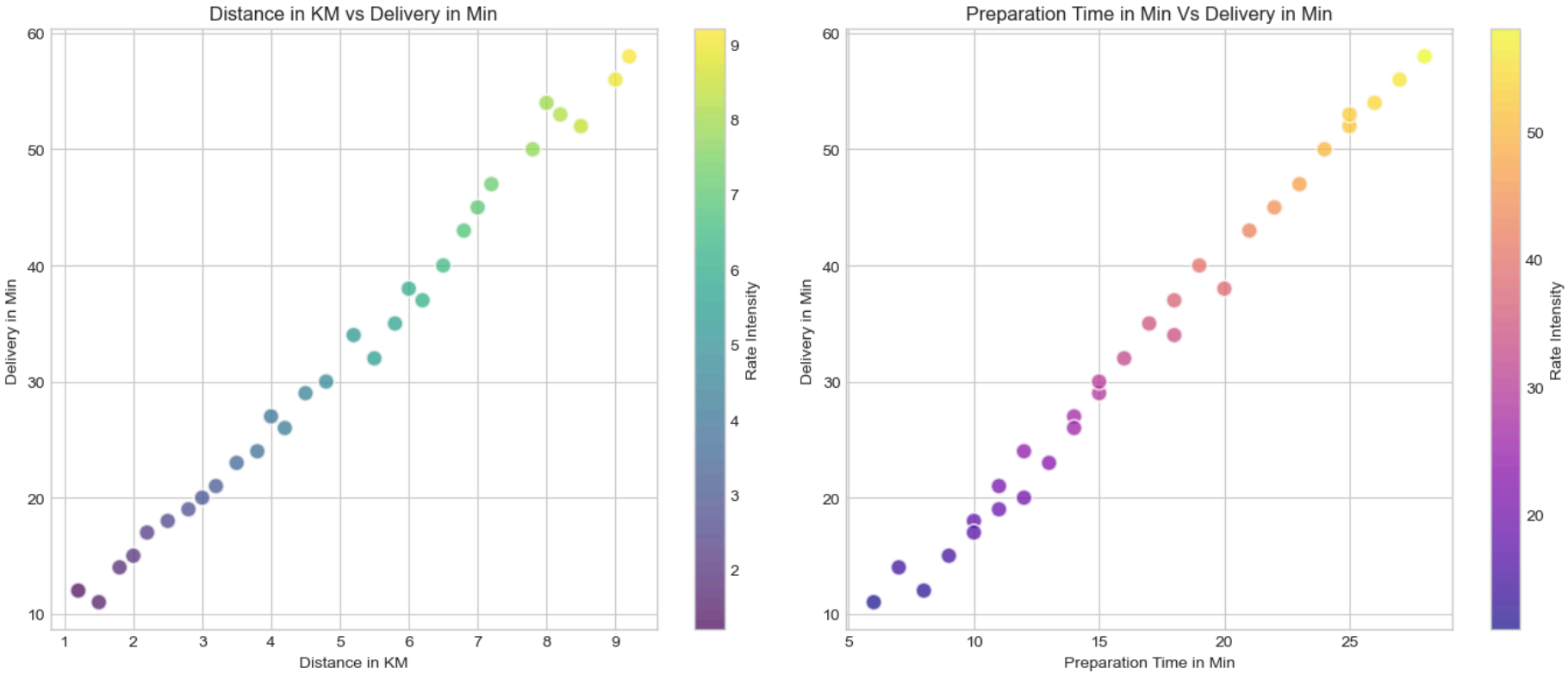
Out[6]:

	distance_km	prep_time_min	delivery_time_min
count	30.000000	30.000000	30.000000
mean	5.063333	16.533333	32.666667
std	2.408818	6.415140	14.465973
min	1.200000	6.000000	11.000000
25%	3.050000	11.250000	20.250000
50%	5.000000	15.500000	31.000000
75%	6.950000	21.750000	44.500000
max	9.200000	28.000000	58.000000

In [10]:
```
# Visualize the data
plt.style.use('seaborn-v0_8-whitegrid')
fig, axis = plt.subplots(1,2, figsize=(14,6))
scatter1 = axis[0].scatter(df['distance_km'], df['delivery_time_min'],
                           c =df['distance_km'], cmap='viridis', s=100, alpha=0.7, edgecolors='white', linewidth=1.5)
axis[0].set_xlabel('Distance in KM')
axis[0].set_ylabel('Delivery in Min')
axis[0].set_title('Distance in KM vs Delivery in Min')
plt.colorbar(scatter1, ax=axis[0], label='Rate Intensity')
```

```
scatter2 = axis[1].scatter(df['prep_time_min'], df['delivery_time_min'],
                           c = df['delivery_time_min'], cmap='plasma', s=100, alpha =0.7, edgecolors='white',linewidth=1.5)
axis[1].set_xlabel('Preparation Time in Min')
axis[1].set_ylabel('Delivery in Min')
axis[1].set_title('Preparation Time in Min Vs Delivery in Min')
plt.colorbar(scatter2, ax=axis[1], label='Rate Intensity')

plt.tight_layout()
plt.show()
```



In [13]:
```
# prepare data for training
X = df[['distance_km','prep_time_min']]
y = df['delivery_time_min']
```

In [15]:
```
# split the data into training and testing data set
random_state =42
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size =0.2)
```

In [16]:
```
# create model and train the model
model = LinearRegression()
model.fit(X_train,y_train)
```

Out[16]:
```
▾ LinearRegression
▶ Parameters
```

In [22]:
```
# check model parameters
distance_coeff = model.coef_[0].round()
prep_time_coeff = model.coef_[1].round()
intercept = model.intercept_
```

In [23]:
```
distance_coeff, prep_time_coeff, intercept
# distance_coeff factor effects delivery time more
```

Out[23]:
```
(np.float64(3.0), np.float64(1.0), np.float64(-0.973524591908415))
```

In [18]:
```
y_pred = model.predict(X_test)
```

In [19]:
```
y_pred
```

Out[19]:
```
array([19.51880307, 51.91680678, 11.23419244, 20.80809586, 52.48594292,
       17.50935365])
```

In [21]:
```
results = pd.DataFrame({'Actual': y_test.values, 'Predicted': y_pred.round()})
results
```

Out[21]:

	Actual	Predicted
0	19	20.0
1	54	52.0
2	12	11.0
3	21	21.0
4	52	52.0
5	18	18.0

In [25]:
```
score = r2_score(y_pred, y_test)
score
```

Out[25]:
```
0.996604571988656
```

In [27]:
```
new_delivery = [[7,15]]
pred_time = model.predict(new_delivery)
pred_time
```

```
C:\Users\G.Guna\desktop\ML\heart_disease_project\env\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does no
t have valid feature names, but LinearRegression was fitted with feature names
  warnings.warn(
```

Out[27]:
```
array([37.22629664])
```

In [31]:
```
print(f"\nQuestion: 7km distance, 15 minutes preparation time")
print(f"Predicted Delivery Time: {pred_time[0]:,.0f} min")
```

```
Question: 7km distance, 15 minutes preparation time
Predicted Delivery Time: 37 min
```

In [ ]: